

RAMESH ADHIKARI

Ph.D. Candidate, School of Computer and Cyber Sciences

Augusta University, Augusta, GA, USA

Email: radhikari@augusta.edu | Web: <https://ramesh-adhikari.github.io> | [Google Scholar](#)

EDUCATION

Ph.D. in Computer and Cyber Sciences (08/2022 – 07/2026 Expected)

Augusta University, Augusta, GA, USA

Advisor: **Dr. Konstantin (Costas) Busch**

M.E. in Computer Engineering (GPA: 3.75/4.0)

Pokhara University, Kathmandu, Nepal, 2018 – 2021

B.E. in Computer Engineering (78.15%)

Tribhuvan University, Kathmandu, Nepal, 2013 – 2017

RESEARCH INTERESTS

Distributed and Parallel Algorithms; Blockchain and Distributed Ledgers; Fault-Tolerant Distributed Systems; Transaction Scheduling; Adversarial Analysis; Fog-Cloud Computing; Cybersecurity.

PROFESSIONAL APPOINTMENTS & EMPLOYMENT

Graduate Research Assistant, Augusta University, USA, 08/2022 – Present

Instructor, Zenlab, Kathmandu, Nepal, 01/2021 – 12/2021

Software Engineer, Sanima Bank, Kathmandu, Nepal 05/2019 – 06/2022

Software Engineer, SmartMobe Solutions Pvt. Ltd, Kathmandu, Nepal 02/2018 – 04/2019

PUBLICATIONS

Conference Proceedings (Peer Reviewed)

- **Ramesh Adhikari**, Costas Busch, and Miroslav Popovic, “On the Efficiency of Dynamic Transaction Scheduling in Blockchain Sharding,” Proceedings of the 39th International Symposium on Distributed Computing (**DISC 2025**), pp 2:1–2:23, Dagstuhl, Germany.
- **Ramesh Adhikari**, Costas Busch, and Dariusz R. Kowalski, “Stable Blockchain Sharding under Adversarial Transaction Generation,” Proceedings of the 36th ACM Symposium on Parallelism in Algorithms and Architectures (**ACM SPAA 2024**), pp. 451-461, Nantes, France.
- **Ramesh Adhikari**, Costas Busch, and Pavan Poudel, “A Poly-Log Approximation for Transaction Scheduling in Fog-Cloud and Beyond,” Proceedings of the 27th International Symposium on Stabilization, Safety, and Security of Distributed Systems (**SSS 2025**), **[Best Student Paper Award]**

- **Ramesh Adhikari**, Costas Busch, and Dariusz R. Kowalski, “Near-Optimal Stable Transaction Processing in Blockchain Sharding,” Proceedings of the 27th International Symposium on Stabilization, Safety, and Security of Distributed Systems (**SSS 2025**), pp 4–20, Kathmandu, Nepal.
- **Ramesh Adhikari** and Costas Busch, “Lockless Blockchain Sharding with Multiversion Control,” Proceedings of the 30th International Colloquium on Structural Information and Communication Complexity (**SIROCCO 2023**), pp. 112 – 131, Alcala De Henares, Spain.

Journal Articles (Published, Peer Reviewed)

- **Ramesh Adhikari**, Costas Busch, Dariusz R. Kowalski, and Abdullah Al-Mamun, “Transaction Processing in Blockchain Sharding: Current Trends and Future Research Directions,” ACM Distributed Ledger Technologies: Research and Practice (**ACM: DLT**), 2025.
- **Ramesh Adhikari** and Suresh Pokharel. “Performance Evaluation of Convolutional Neural Networks Using Synthetic Medical Data Augmentation Generated by GANs,” International Journal of Image and Graphics, 2023.
- **Ramesh Adhikari**, Costas Busch, and Pavan Poudel, “A Poly-Log Approximation for Transaction Scheduling in Fog–Cloud and Beyond,” Theoretical Computer Science (**TCS**), 2026.

Journal Articles (Under Review / Revision)

- **Ramesh Adhikari**, Costas Busch, and Dariusz R. Kowalski, “Stable Blockchain Sharding under Adversarial Transaction Generation,” Distributed Computing (**DC Springer**), *First Round Revision*.
- **Ramesh Adhikari**, Costas Busch, and Miroslav Popovic, “On the Efficiency of Dynamic Transaction Scheduling in Blockchain Sharding,” ACM Transactions on Parallel Computing (**ACM TOPC**).
- **Ramesh Adhikari**, Costas Busch, and Dariusz R. Kowalski, “Near-Optimal Stable Transaction Processing in Blockchain Sharding,” Theoretical Computer Science (**TCS**), *Invited for Special Issues*.

Preprints

- **Ramesh Adhikari**, Costas Busch, and Miroslav Popovic, “Fast Transaction Scheduling in Blockchain Sharding,” arXiv preprint, 2024, <https://arxiv.org/abs/2405.15015>

GRANTS AND FUNDING

- Contributed to the preparation of **two NSF research grant proposals (CSR, AF)**
 - PhD Supported by NSF Grant #2131538.
 - Multiple NSF and university travel awards (SPAA, DISC, CANS).
 - **Planned submissions:** NSF CAREER, NSF SaTC / CSR.
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TEACHING EXPERIENCE

Guest Lecturer – CSCI 8250 Quantum Computing (Graduate Level)

Augusta University, Augusta, GA, USA, Spring 2026

Topics: quantum programming, OpenQASM, Qiskit, IBM Quantum hands-on circuit execution.

Teaching Interests: Data Structures and Algorithms; Distributed Systems; Blockchain; Databases; Parallel Computing; Cybersecurity; Cloud Computing; Advanced Topics in Computer Security.

STUDENT SUPERVISION (Selected)

- Sushant Dahal, B.Sc. Thesis (Co-Supervisor), Thapar Institute of Engineering and Technology, 2025
 - Dipesh Bhattarai, B.Sc. Thesis (Co-Supervisor), Thapar Institute of Engineering and Technology, 2025
 - Dhiraj Sharma, M.S. Thesis (Co-Supervisor), Pokhara University, 2023
 - Mentored undergraduate student teams for the EVOLVE 2.0 software competition, Kathford International College, 2017
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SELECTED AWARDS & HONORS

- **Best Student Paper Award, \$585, SSS'25** (2025).
 - Full Tuition Waiver and Graduate Research Assistantship, Augusta University (2022–present).
 - Full Scholarship for M.E. in Computer Engineering, Pokhara University (2018).
 - **NSF Student Travel Awards** (Multiple) totaling **\$2,090** (SPAA 2024, DISC 2022, CANS 2023).
 - **University Travel Awards** (Multiple) totaling **\$2,000**, Augusta University (2023–2024).
 - Star Performance Excellence Award, Sanima Bank Ltd. (2021).
 - First Runner-Up in a software competition at Kantipur Engineering, Tribhuvan University (2015).
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TALKS & PRESENTATIONS

Invited Talks

- Stable Blockchain Sharding under Adversarial Transaction Generation, EECS Department, Syracuse University, USA, Nov 2024.
- Stable Blockchain Sharding under Adversarial Transaction Generation, School of Computer and Cyber Sciences, Augusta University, USA, Nov 2024.

Conference Presentations

- Stable Blockchain Sharding under Adversarial Transaction Generation, **SPAA 2024**, Nantes, France.
- Lockless Blockchain Sharding with Multiversion Control, **SIROCCO 2023**, Alcalá de Henares, Spain.
- Near-Optimal Stability for Distributed Transaction Processing, **SSS 2025**, Kathmandu, Nepal.

Poster Presentations

- Augusta University Research Day, March 2025; Cyber-Physical Systems Innovation Symposium, September 2025; Emerging Data Science Workshop, March 2023.
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ACADEMIC SERVICE

Registration Chair, SSS 2025

Web Chair, NSF Innovation Engine Symposium on Cyber-Security and Cyber-Physical Systems (2025)

Reviewer (Journals): IEEE TGCN; IEEE TMC; Journal of Network and Computer Applications

Reviewer (Conferences): DISC'25, PODC'24, SIROCCO'24, IEEE Blockchain'23 & Blockchain'24

INDUSTRY EXPERIENCE (Selected)

Software Engineer, Sanima Bank, Nepal (2019–2022)

Led development of secure, scalable banking applications; integrated APIs with core banking systems

Software Engineer, SmartMobe Solutions, Nepal (2018–2019)

Developed APIs for mobile and web platforms; integrated payment gateways

TECHNICAL SKILLS

Programming: C, C++, Python, Go, Java

Distributed & Parallel: MPI, OpenMP, CUDA

Systems & Tools: Linux, Git, AWS, LaTeX
